

Anirudh S Shirahati

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SUMMARY

Senior Software Engineer with 6+ years of experience building Python based backend systems, REST APIs, cloud native applications and enterprise software solutions. Experienced delivering production applications across Azure and Google Cloud Platform with expertise in backend development, data driven systems, automation and CI/CD practices. Focused on Applied AI and LLM application engineering, designing and building AI powered software solutions using modern AI frameworks, cloud platforms and software engineering practices. Hands-on experience developing end to end AI applications involving large language models, document intelligence, conversational workflows, structured outputs, local model inference and cloud deployment.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript

AI Engineering: LLM Application Development, AI Agents, Retrieval Augmented Generation (RAG), Prompt Engineering, LangChain, LiteLLM, Hugging Face Transformers, Chroma, Vector Stores, Structured Outputs, JSON Schema, Model Quantization, Claude, OpenRouter

Backend & APIs: FastAPI, Flask, REST APIs, Pydantic, Backend Services

Cloud & DevOps: Microsoft Azure, Google Cloud Platform (GCP), Docker, GitHub Actions, Azure DevOps, CI/CD

Data & Databases: SQL, MySQL, MongoDB, Pandas, NumPy, Data Processing, Data Visualization

Frontend: Next.js, Vue.js, HTML, CSS

Developer Tools: Git, GitHub, Claude Code, GitHub Copilot, Cursor, Visual Studio Code

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Independent AI Engineering & Software Development | June 2025 - Present

- Designed and built AI powered software applications while applying modern AI engineering practices across LLM powered workflows, backend systems and cloud deployments.
- Engineered end to end AI applications using Python, FastAPI, Docker and cloud deployment workflows, integrating large language models into practical business workflows.
- Implemented LLM application patterns including Retrieval Augmented Generation (RAG), agent workflows, vector based retrieval, structured generation and local model inference.
- Built full stack applications with backend APIs, authentication, data persistence, document processing workflows and automated deployment pipelines.
- Leveraged AI assisted software development tools including Claude Code, GitHub Copilot, Cursor and VS Code to improve development velocity, debugging and iterative implementation.
- Designed AI applications with production focused engineering practices including API architecture, validation workflows, automated testing and cloud deployment.

Manager, Solar PV Systems (Software Engineering) | RWE Clean Energy Services, LLC | June 2021 - June 2025

- Designed and delivered Python based applications, backend services and internal platforms supporting engineering, operational and financial business workflows.
- Built REST APIs and backend services using Python and FastAPI to integrate business systems and automate operational processes.
- Developed interactive Dash applications that transformed operational and engineering data into analytics platforms supporting reporting, monitoring and business decision making.
- Automated financial and operational modeling workflows, reducing manual effort by approximately 70% and improving reporting efficiency.
- Optimized SQL queries and database interactions to improve application performance and support scalable data processing.
- Built and deployed containerized applications using Docker and Microsoft Azure, supported by CI/CD pipelines implemented with Azure DevOps and GitHub Actions.
- Owned software features from requirements through deployment, collaborating closely with engineers, business users and stakeholders to deliver production ready solutions.
- Mentored developers through code reviews and technical collaboration while promoting maintainable software development practices.

Research Assistant (Cloud Software Engineer) | UNCC | August 2020 - May 2021

- Developed and deployed containerized Python applications using Google Cloud Platform services including Cloud Run, BigQuery and Cloud Storage.
- Built backend services and APIs supporting machine learning experimentation, data processing and research workflows.
- Automated cloud environment setup and application deployment processes to improve development efficiency.
- Collaborated with researchers and engineering teams to support cloud based analytics and software initiatives.

Software Developer / Java Developer | Multiple Clients | February 2016 - April 2019

- Developed backend modules and enterprise applications for banking and financial services systems using Java and Spring Framework.
- Designed and implemented backend services supporting customer facing and internal business operations.
- Improved application performance by up to 30% through code optimization and system enhancements.
- Collaborated with Agile development teams throughout the software development lifecycle.

SELECTED AI ENGINEERING PROJECTS

Inkwell - LLM Powered Document Automation | Python, Transformers, Llama 3.1, Hugging Face, OpenCV, Pydantic, Gradio, GCP

- Built an LLM powered document automation application that converts natural language instructions into accurately positioned data on scanned, non interactive PDF forms.
- Engineered a computer vision pipeline using OpenCV morphological analysis to detect and recover 227 addressable form cells from flat scanned documents without relying on existing form fields.
- Implemented local inference using Meta Llama 3.1 8B Instruct through Hugging Face Transformers with 4 bit NF4 quantization using bitsandbytes, enabling execution within approximately 5.6 GB VRAM.
- Designed JSON Schema constrained generation and Pydantic validation workflows to enforce structured model outputs and enable automated correction of invalid responses.
- Built automated verification tests ensuring document integrity and confirming protected regions remain pixel identical to the source.
- Developed a privacy focused architecture that eliminates external API dependencies and keeps sensitive document processing within the local execution environment.

PreLegal - AI Powered Legal Document Drafting Platform | Python, FastAPI, Next.js, Claude, OpenRouter, Docker, GitHub Actions

- Built and deployed a full stack SaaS platform that uses conversational AI to generate professional legal agreements from user provided requirements.
- Implemented conversational AI workflows that gather user requirements, generate structured document content and produce customized legal agreements.
- Developed backend services using FastAPI with user authentication, document persistence and PDF generation capabilities.
- Integrated Claude through OpenRouter as the conversational drafting engine supporting 11 legal document types based on Common Paper open standards.
- Created automated CI/CD workflows using GitHub Actions to deploy the application to Render.

EDUCATION

Master of Science - Computer Science (Data Science)

University of North Carolina at Charlotte

Bachelor of Engineering - Computer Engineering

University of Mumbai